Anjaney Malhotra

Noida, Uttar Pradesh • anjaneymalhotra@gmail.com • +91 9599366178
GitHub • LinkedIn • **Portfolio:** anjaney336.github.io/portfolio

SCAN PORTFOLIO

May 2027 internships | Quantitative Research / Developer Internships | Across India

EDUCATION

Vellore Institute of Technology (VIT), Vellore *2024–2028 (Expected)*

B.Tech in Mechanical Engineering, CGPA: **8.86 / 10**

EXPERIENCE

Alvionx *Sep 2026–Present*

Fintech Intern

- Journaled **172 completed trades** across backtesting work; documented liquidation-based ETH/BTC signals, entries, exits and trade rationale.
- Backtested Reliance price triggers: invest **10% of remaining cash** per 5% decline from the last buy; exit 5% above weighted-average cost.

LearnTube.ai *Oct 2025–Jan 2026*

Data Analytics Intern

- Developed forecasting and data-validation workflows using stationarity tests, seasonal decomposition, error comparisons and checks for missing values, duplicates and outliers.

PROJECTS

Systematic Strategy Research *Code & evidence*

Python, NumPy | Chronological selection and portfolio accounting

- Built **9 rule variants** with rolling 504/126-observation selection, volatility sizing, delayed signals and drift-aware cost accounting; added **12 tests**.
- Evaluated **1,422 historical observations**, five fee scenarios and block-bootstrap comparisons; documented underperformance versus buy-and-hold.

Derivatives Hedging Research *Code & evidence*

Python | Greeks, nonlinear risk and self-financing hedges

- Implemented pricing, implied volatility and delta/gamma hedges with financing, costs and position caps; validated Greeks and accounting with **26 tests**.
- Evaluated **900 hedge runs plus 600 delta/gamma comparisons**; published hedge-error distributions under diffusion/jump assumptions using model-priced options.

Integrated Real Assets Research *Code & evidence*

Python | Listed-property signals, tail risk and execution

- Integrated cointegration/Kalman signals, lagged expected-shortfall scaling and execution stress; evaluated **1,675 historical observations** with **53 tests**.
- Published negative baseline and overlay returns, cost sensitivities and **200 paired-policy execution runs**; distinguished simulation from market evidence.

Opti-Workbench — Constrained Portfolio Optimisation *Code & evidence*

Python, CVXPY | Synthetic allocation research

- Implemented empirical-CVaR convex seeding and PSO/GA refinement with allocation, budget and cardinality constraints; documented objective differences.
- Added train-only rolling evaluation with drift-adjusted holdings and turnover costs; compared four solvers with an equal-weight reference on synthetic data.

SKILLS

Quantitative: Time series, backtesting, option Greeks, portfolio risk, execution and inference.

Tools: Python, pandas, NumPy, SQL/SQLite, Excel, scikit-learn, statsmodels, Git/GitHub.

CERTIFICATIONS

Google AI Essentials — Google / Coursera (Dec 2025) | MATLAB Onramp — MathWorks (Jan 2026) | Python — Kaggle (Dec 2025) | Data Analytics — Skillsetmaster (Jan 2026)

Artificial Intelligence — Skillumni (Apr 2026) | Onshape Hands-On Test Drive — Onshape by PTC (Aug 2026) | Data Analytics Assessment — LearnTube.ai (Jan 2026, ID DJA-B-1-1989388-0) | Time Series Forecasting Assessment — LearnTube.ai (Jan 2026, ID DJA-B-1-1989425-0)